

CHATHAM 1 SE, MI, USA

WMO: 725380

Lat: 46.3345N

Lon: 86.9200W

Elev: 267

StdP: 98.16

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.4	-20.4	-27.4	0.3	-23.5	-24.5	0.4	-20.4	6.9	-9.1	6.3	-8.6	1.1	N/A	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	29.0	21.3	27.3	20.3	25.7	19.1	22.4	26.8	21.3	25.5	20.2	24.2	1.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	16.0	24.7	19.8	15.0	23.7	18.8	14.1	22.5	67.0	26.7	63.0	25.7	59.3	24.2	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.8	4.1	DB	-28.7	31.7	4.2	1.6	-31.8	32.9	-34.2	33.8	-36.6	34.7	-39.7	35.9	(4)
(5)			WB	-29.8	24.8	4.5	1.3	-33.1	25.7	-35.7	26.5	-38.3	27.3	-41.6	28.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-8.5	-8.8	-3.3	3.1	10.0	15.1	18.2	17.4	13.8	7.2	0.5	-5.8
	DBStd	10.85	5.96	6.17	6.34	4.93	4.94	3.97	3.56	3.28	4.37	4.69	5.19	5.40
	HDD10.0	2703	573	527	416	213	63	6	0	0	13	117	287	489
	HDD18.3	4977	831	761	672	456	262	113	47	58	147	348	536	747
	CDD10.0	871	0	0	2	7	63	158	254	229	127	29	2	0
	CDD18.3	103	0	0	0	0	4	15	44	28	11	1	0	0
	CDH23.3	993	0	0	1	1	57	161	434	252	81	6	0	0
	CDH26.7	203	0	0	0	0	9	29	107	44	13	0	0	0
Wind	WSAvg	1.8	2.2	2.2	2.1	2.2	1.9	1.4	1.2	1.2	1.3	1.6	1.9	2.0
Precipitation	PrecAvg	809	46	34	45	50	71	76	81	83	103	92	78	52
	PrecMax	1060	131	106	120	112	178	170	175	204	191	202	189	130
	PrecMin	570	11	2	9	9	4	16	18	10	22	27	6	11
	PrecStd	136	25	18	26	25	37	38	44	38	40	40	35	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	6.7	17.5	21.7	27.8	29.5	31.6	30.2	28.8	23.9	16.3	7.2
		MCWB	2.5	3.7	11.3	11.9	17.8	21.5	23.4	22.0	21.0	16.0	11.0	5.3
	2%	DB	2.6	3.7	11.0	17.6	24.9	27.2	29.5	27.8	25.6	20.0	11.9	4.2
		MCWB	1.4	1.7	8.3	10.8	16.7	19.9	21.6	20.6	19.7	14.5	8.8	3.3
	5%	DB	1.2	1.2	8.0	14.6	22.0	25.1	27.7	26.1	23.5	17.4	9.8	2.7
		MCWB	0.4	-0.4	5.0	7.7	15.0	18.2	21.0	19.2	18.3	13.7	7.5	1.8
	10%	DB	0.0	-0.2	5.3	11.8	19.2	23.3	25.7	24.5	21.5	14.8	7.8	1.3
		MCWB	-0.8	-1.3	2.8	6.5	13.6	17.2	19.6	18.7	17.5	12.2	6.2	0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	5.0	15.1	13.0	20.1	22.8	24.7	23.2	22.5	17.2	12.1	7.0
		MCDB	3.7	7.2	20.9	17.4	24.9	27.7	29.3	27.9	26.6	20.0	15.0	7.6
	2%	WB	1.4	2.1	9.1	10.2	18.3	20.9	22.8	21.7	20.8	15.5	10.0	3.8
		MCDB	2.3	4.0	11.0	13.8	22.3	25.6	27.2	25.8	23.8	18.5	11.9	4.4
	5%	WB	0.4	0.1	5.1	7.5	16.5	19.2	21.6	20.5	19.6	13.9	7.6	2.2
		MCDB	1.1	1.3	7.2	12.3	20.6	22.9	26.1	24.4	22.1	16.4	9.4	2.7
	10%	WB	-1.0	-1.2	2.3	5.6	14.8	17.8	20.5	19.4	18.1	12.2	6.0	0.8
		MCDB	-0.3	0.0	4.2	9.8	18.5	21.7	24.8	22.9	20.5	14.5	7.6	1.5
Mean Daily Temperature Range	5% DB	MDBR	7.5	9.6	10.0	10.3	12.1	12.2	12.3	12.0	11.3	8.9	7.3	6.9
		MCDBR	7.2	10.1	12.6	16.5	16.1	14.8	14.3	14.3	13.3	12.4	10.7	7.3
		MCWBR	6.6	8.1	8.5	10.4	9.2	8.2	7.8	8.2	7.3	8.2	7.5	6.7
	5% WB	MCDBR	7.2	9.3	10.9	14.4	13.2	12.5	12.7	12.1	11.0	11.3	9.7	6.9
		MCWBR	6.5	7.6	7.8	9.8	8.6	7.8	7.6	7.3	7.3	8.2	7.1	6.3
Clear-Sky Solar Irradiance	taub	0.264	0.276	0.310	0.340	0.370	0.376	0.382	0.369	0.354	0.327	0.298	0.266	
	taud	2.448	2.396	2.393	2.396	2.377	2.391	2.394	2.429	2.473	2.530	2.532	2.493	
	Ebn at Noon	854	908	921	918	896	889	879	878	862	839	804	813	
	Edh at Noon	66	86	101	110	117	116	114	106	93	76	62	58	
All-Sky Solar Radiation	RadAvg	1.10	1.95	3.37	4.47	5.47	5.92	6.05	5.12	3.80	2.26	1.21	0.85	
	RadStd	0.09	0.26	0.31	0.46	0.46	0.40	0.45	0.30	0.33	0.26	0.26	0.15	0.07

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page